LAB 06

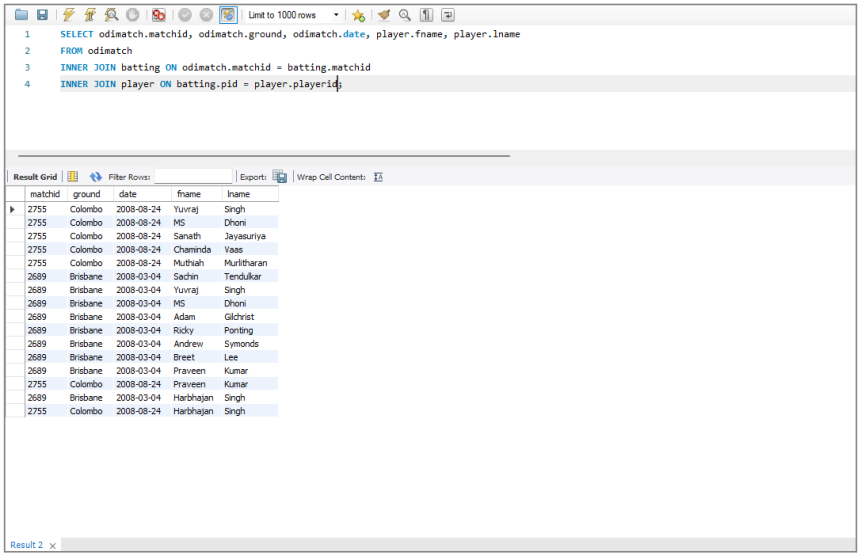
Implementation of Various Join Operations

Aim: Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

1. [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

Query:

```
SELECT odimatch.matchid, odimatch.ground, odimatch.date, player.fname, player.lname
FROM odimatch
INNER JOIN batting ON odimatch.matchid = batting.matchid
INNER JOIN player ON batting.pid = player.playerid;
```

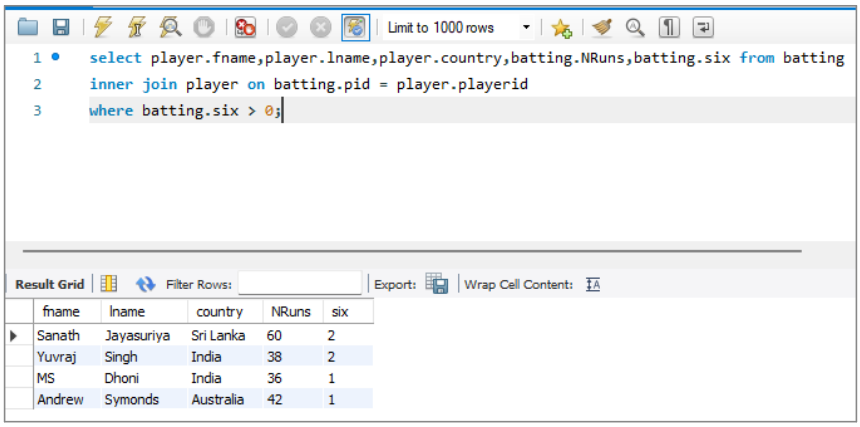


matchid	ground	date	fname	lname
2755	Colombo	2008-08-24	Yuvraj	Singh
2755	Colombo	2008-08-24	MS	Dhoni
2755	Colombo	2008-08-24	Sanath	Jayasuriya
2755	Colombo	2008-08-24	Chaminda	Vaas
2755	Colombo	2008-08-24	Muthiah	Murlitharan
2689	Brisbane	2008-03-04	Sachin	Tendulkar
2689	Brisbane	2008-03-04	Yuvraj	Singh
2689	Brisbane	2008-03-04	MS	Dhoni
2689	Brisbane	2008-03-04	Adam	Gilchrist
2689	Brisbane	2008-03-04	Ricky	Ponting
2689	Brisbane	2008-03-04	Andrew	Symonds
2689	Brisbane	2008-03-04	Breet	Lee
2689	Brisbane	2008-03-04	Praveen	Kumar
2755	Colombo	2008-08-24	Praveen	Kumar
2689	Brisbane	2008-03-04	Harbhajan	Singh
2755	Colombo	2008-08-24	Harbhajan	Singh

2. [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

Query:

```
select player.fname,player.lname,player.country,batting.NRuns,batting.six from batting
inner join player on batting.pid = player.playerid
where batting.six > 0;
```

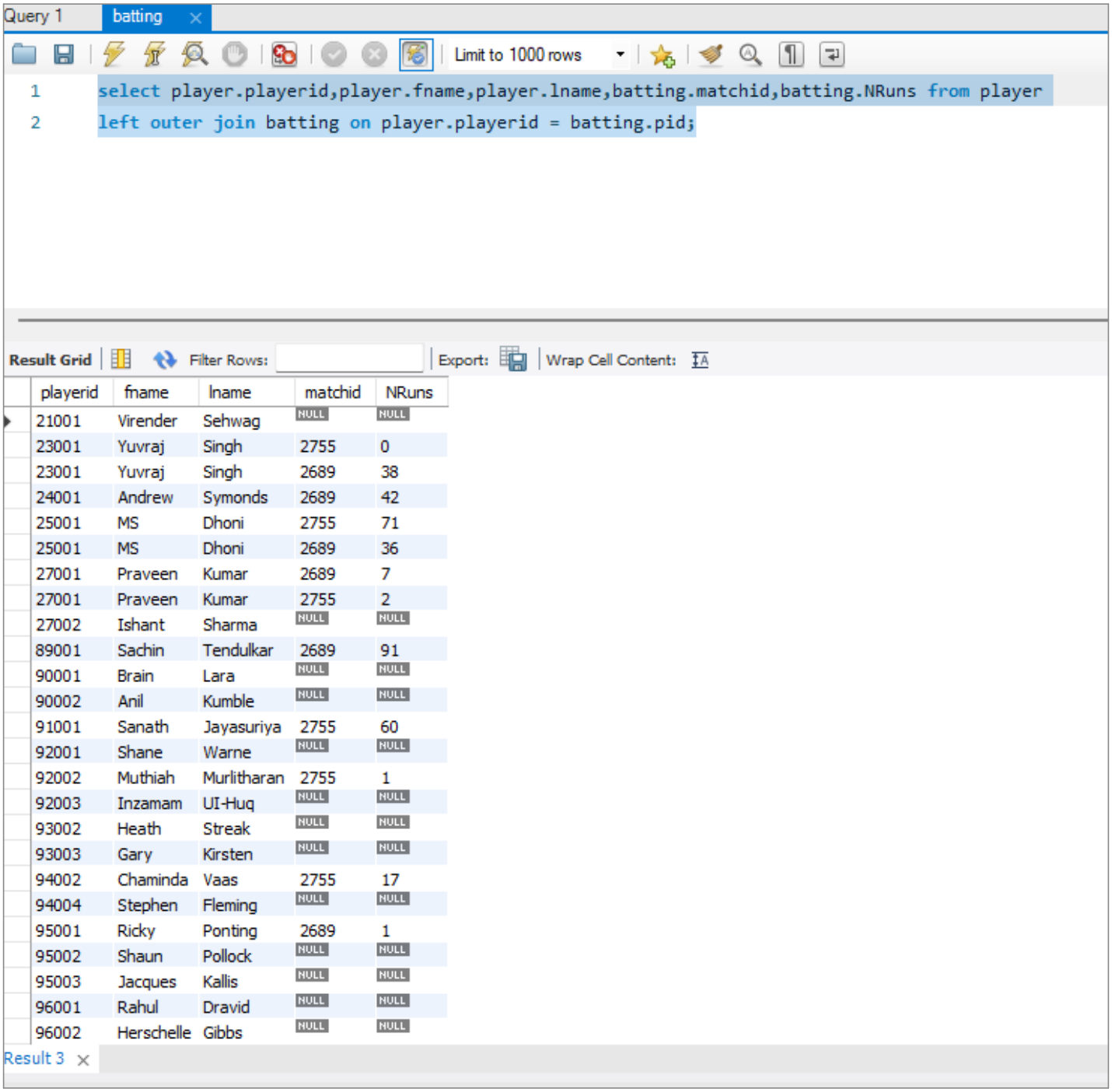


fname	lname	country	NRuns	six
Sanath	Jayasuriya	Sri Lanka	60	2
Yuvraj	Singh	India	38	2
MS	Dhoni	India	36	1
Andrew	Symonds	Australia	42	1

3. [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

Query:

select player.playerid,player.fname,player.lname,batting.matchid,batting.NRuns from player
left outer join batting on player.playerid = batting.pid;

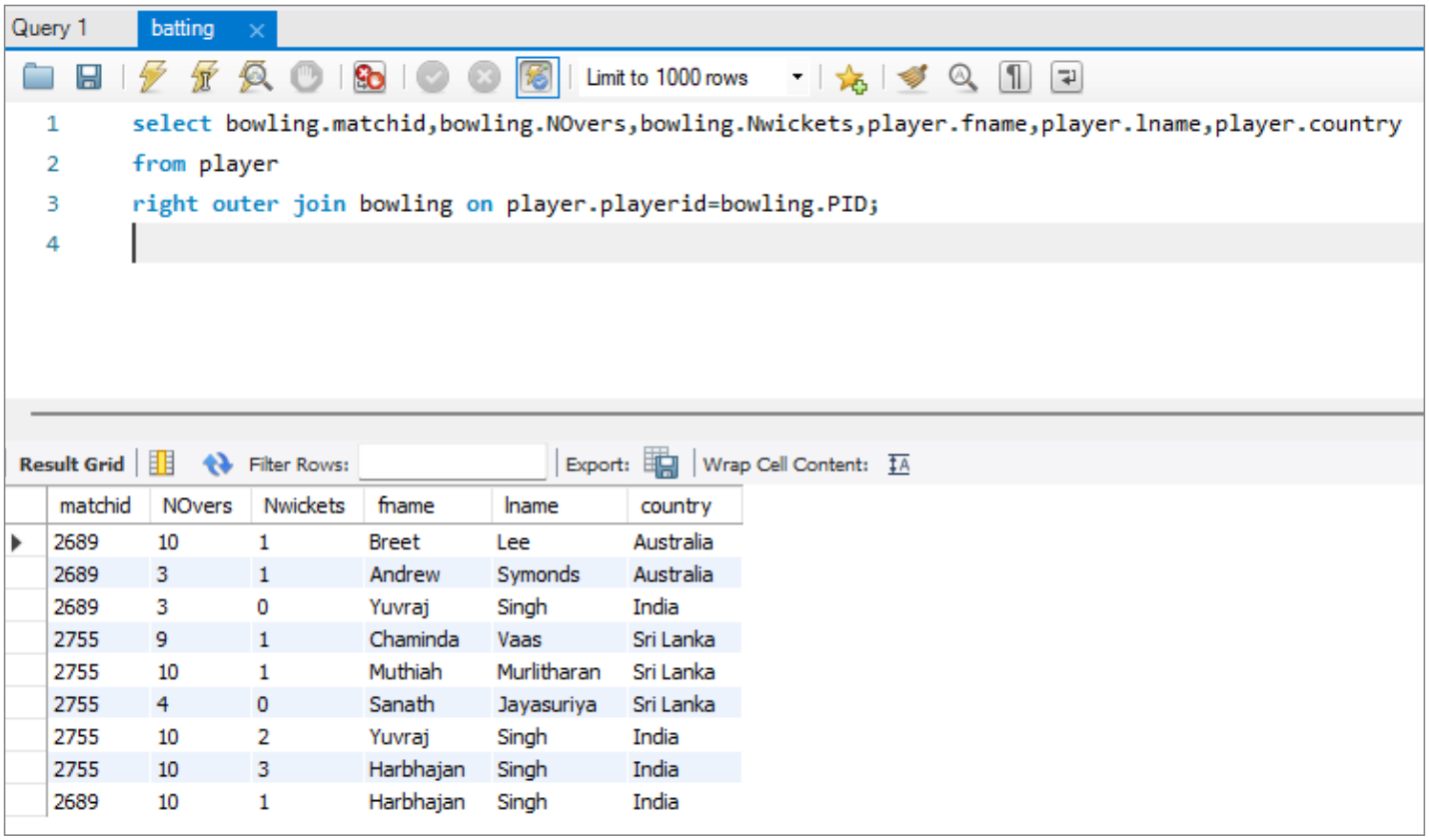


playerid	fname	lname	matchid	NRuns
21001	Virender	Sehwag	NULL	NULL
23001	Yuvraj	Singh	2755	0
23001	Yuvraj	Singh	2689	38
24001	Andrew	Symonds	2689	42
25001	MS	Dhoni	2755	71
25001	MS	Dhoni	2689	36
27001	Praveen	Kumar	2689	7
27001	Praveen	Kumar	2755	2
27002	Ishant	Sharma	NULL	NULL
89001	Sachin	Tendulkar	2689	91
90001	Brain	Lara	NULL	NULL
90002	Anil	Kumble	NULL	NULL
91001	Sanath	Jayasuriya	2755	60
92001	Shane	Warne	NULL	NULL
92002	Muthiah	Murlitharan	2755	1
92003	Inzamam	UI-Huq	NULL	NULL
93002	Heath	Streak	NULL	NULL
93003	Gary	Kirsten	NULL	NULL
94002	Chaminda	Vaas	2755	17
94004	Stephen	Fleming	NULL	NULL
95001	Ricky	Ponting	2689	1
95002	Shaun	Pollock	NULL	NULL
95003	Jacques	Kallis	NULL	NULL
96001	Rahul	Dravid	NULL	NULL
96002	Herschelle	Gibbs	NULL	NULL

4. [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

Query:

```
select bowling.matchid,bowling.NOvers,bowling.Nwickets,player.fname,player.lname,player.country
from player
right outer join bowling on player.playerid=bowling.PID;
```

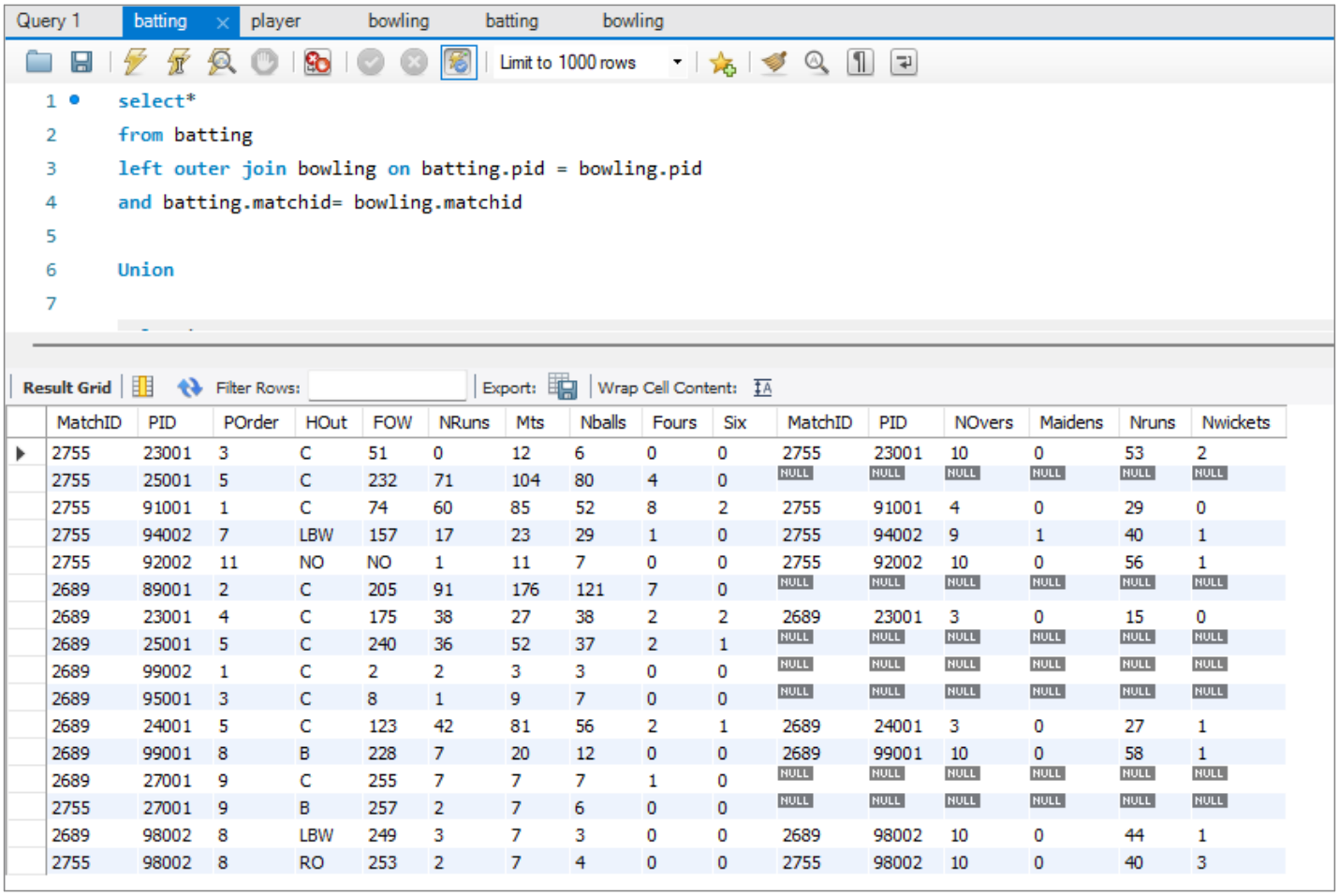


matchid	NOvers	Nwickets	fname	lname	country
2689	10	1	Breet	Lee	Australia
2689	3	1	Andrew	Symonds	Australia
2689	3	0	Yuvraj	Singh	India
2755	9	1	Chaminda	Vaas	Sri Lanka
2755	10	1	Muthiah	Murlitharan	Sri Lanka
2755	4	0	Sanath	Jayasuriya	Sri Lanka
2755	10	2	Yuvraj	Singh	India
2755	10	3	Harbhajan	Singh	India
2689	10	1	Harbhajan	Singh	India

5. [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

Query:

```
select *
from batting
left outer join bowling on batting.pid = bowling.pid
and batting.matchid = bowling.matchid
Union
select *
from batting
right outer join bowling on batting.pid = bowling.pid
and batting.matchid = bowling.matchid;
```

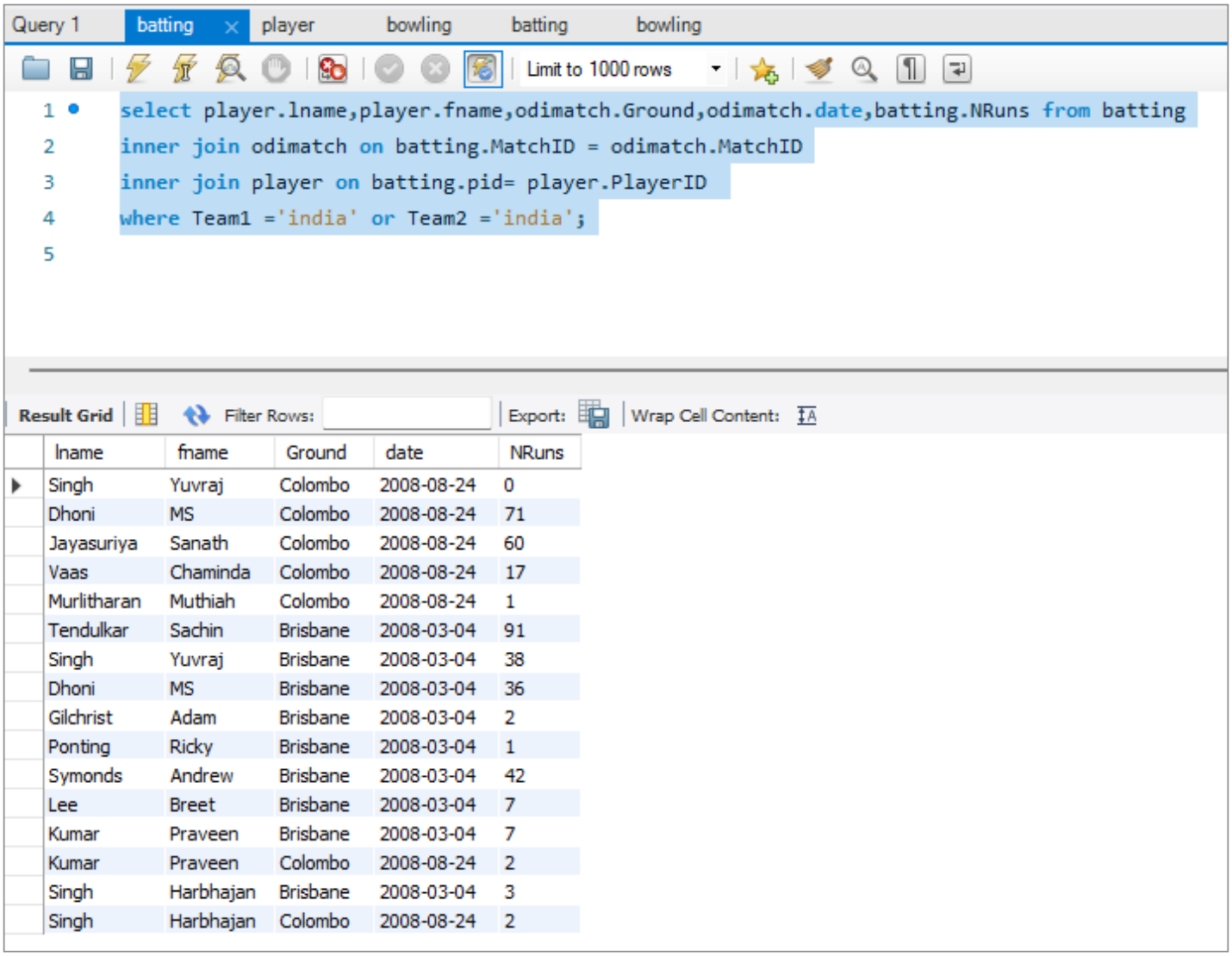


MatchID	PID	POrder	HOut	FOW	NRuns	Mts	Nballs	Fours	Six	MatchID	PID	NOvers	Maidens	Nruns	Nwickets
2755	23001	3	C	51	0	12	6	0	0	2755	23001	10	0	53	2
2755	25001	5	C	232	71	104	80	4	0	NULL	NULL	NULL	NULL	NULL	NULL
2755	91001	1	C	74	60	85	52	8	2	2755	91001	4	0	29	0
2755	94002	7	LBW	157	17	23	29	1	0	2755	94002	9	1	40	1
2755	92002	11	NO	NO	1	11	7	0	0	2755	92002	10	0	56	1
2689	89001	2	C	205	91	176	121	7	0	NULL	NULL	NULL	NULL	NULL	NULL
2689	23001	4	C	175	38	27	38	2	2	2689	23001	3	0	15	0
2689	25001	5	C	240	36	52	37	2	1	NULL	NULL	NULL	NULL	NULL	NULL
2689	99002	1	C	2	2	3	3	0	0	NULL	NULL	NULL	NULL	NULL	NULL
2689	95001	3	C	8	1	9	7	0	0	NULL	NULL	NULL	NULL	NULL	NULL
2689	24001	5	C	123	42	81	56	2	1	2689	24001	3	0	27	1
2689	99001	8	B	228	7	20	12	0	0	2689	99001	10	0	58	1
2689	27001	9	C	255	7	7	7	1	0	NULL	NULL	NULL	NULL	NULL	NULL
2755	27001	9	B	257	2	7	6	0	0	NULL	NULL	NULL	NULL	NULL	NULL
2689	98002	8	LBW	249	3	7	3	0	0	2689	98002	10	0	44	1
2755	98002	8	RO	253	2	7	4	0	0	2755	98002	10	0	40	3

6. [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

Query:

```
select player.lname,player.fname,odimatch.Ground,odimatch.date,batting.NRuns from batting
inner join odimatch on batting.MatchID = odimatch.MatchID
inner join player on batting.pid = player.PlayerID
where Team1 ='india' or Team2 ='india';
```

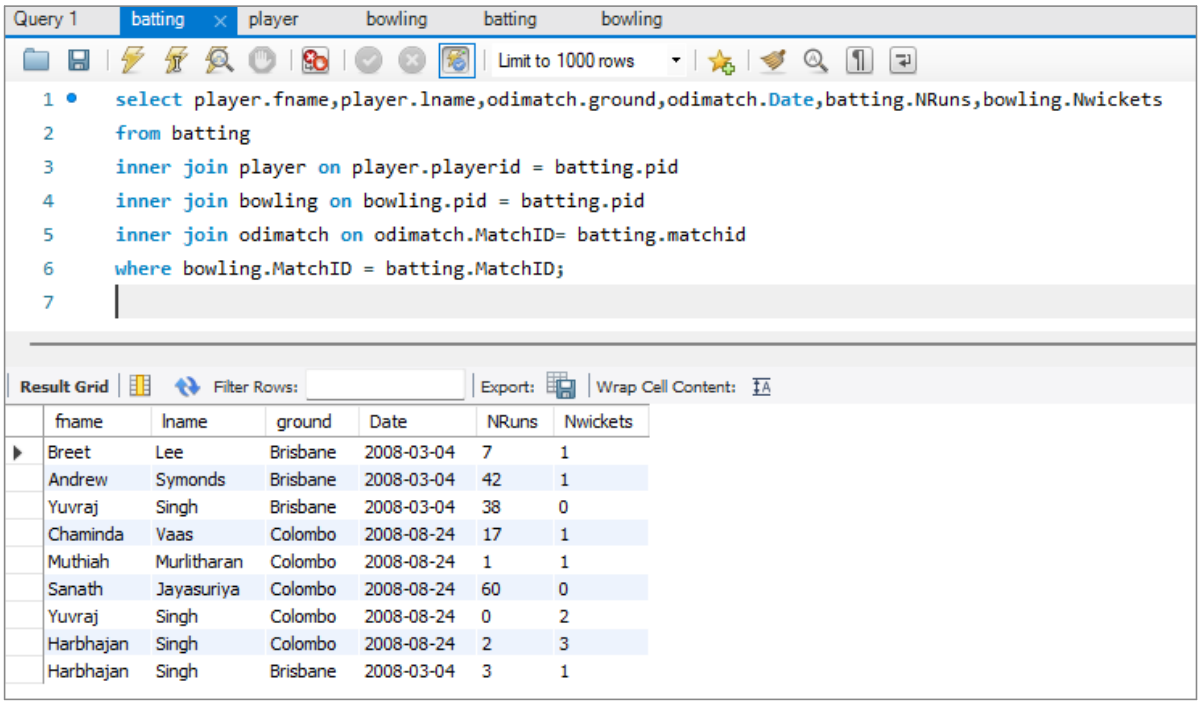


lname	fname	Ground	date	NRuns
Singh	Yuvraj	Colombo	2008-08-24	0
Dhoni	MS	Colombo	2008-08-24	71
Jayasuriya	Sanath	Colombo	2008-08-24	60
Vaas	Chaminda	Colombo	2008-08-24	17
Murlitharan	Muthiah	Colombo	2008-08-24	1
Tendulkar	Sachin	Brisbane	2008-03-04	91
Singh	Yuvraj	Brisbane	2008-03-04	38
Dhoni	MS	Brisbane	2008-03-04	36
Gilchrist	Adam	Brisbane	2008-03-04	2
Ponting	Ricky	Brisbane	2008-03-04	1
Symonds	Andrew	Brisbane	2008-03-04	42
Lee	Breet	Brisbane	2008-03-04	7
Kumar	Praveen	Brisbane	2008-03-04	7
Kumar	Praveen	Colombo	2008-08-24	2
Singh	Harbhajan	Brisbane	2008-03-04	3
Singh	Harbhajan	Colombo	2008-08-24	2

7. [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

Query:

```
select player.fname,player.lname,odimatch.ground,odimatch.Date,batting.NRuns,bowling.Nwickets
from batting
inner join player on player.playerid = batting.pid
inner join bowling on bowling.pid = batting.pid
inner join odimatch on odimatch.MatchID = batting.matchid
where bowling.MatchID = batting.MatchID;
```

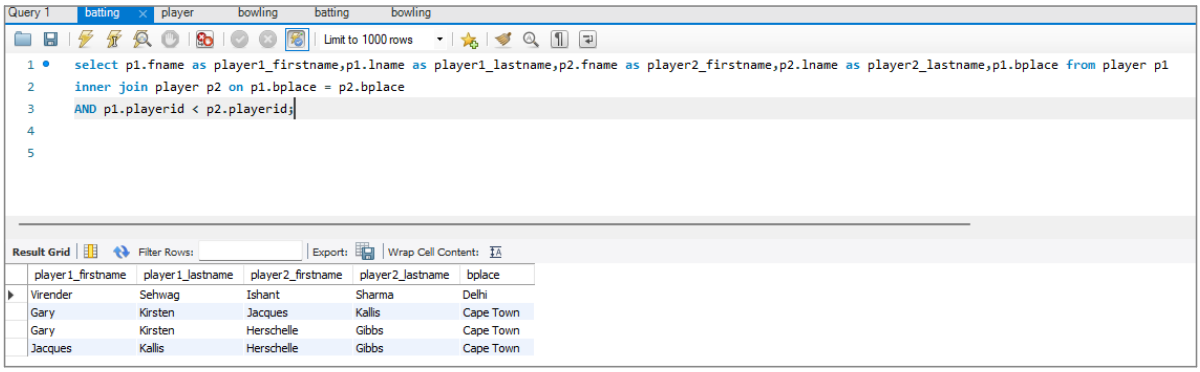


fname	lname	ground	Date	NRuns	Nwickets
Breet	Lee	Brisbane	2008-03-04	7	1
Andrew	Symonds	Brisbane	2008-03-04	42	1
Yuvraj	Singh	Brisbane	2008-03-04	38	0
Chaminda	Vaas	Colombo	2008-08-24	17	1
Muthiah	Murlitharan	Colombo	2008-08-24	1	1
Sanath	Jayasuriya	Colombo	2008-08-24	60	0
Yuvraj	Singh	Colombo	2008-08-24	0	2
Harbhajan	Singh	Colombo	2008-08-24	2	3
Harbhajan	Singh	Brisbane	2008-03-04	3	1

8. [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

Query:

```
select p1.fname as player1_firstname,p1.lname as player1_lastname,p2.fname as player2_firstname,p2.lname as
player2_lastname,p1.bplace from player p1
inner join player p2 on p1.bplace = p2.bplace
AND p1.playerid < p2.playerid;
```

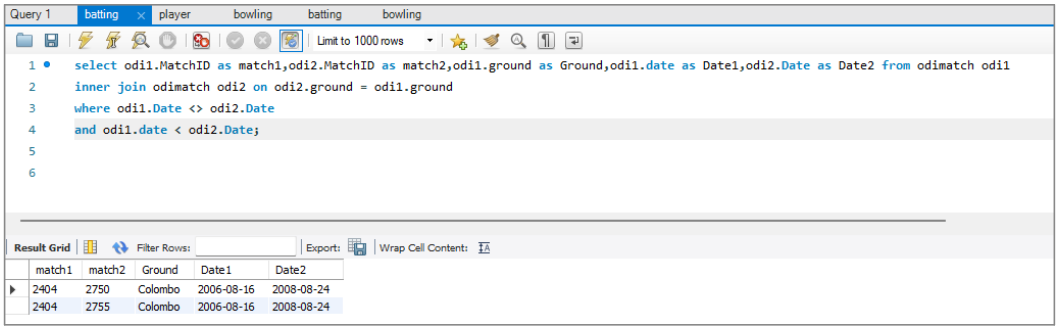


player1_firstname	player1_lastname	player2_firstname	player2_lastname	bplace
Virender	Sehwag	Ishant	Sharma	Delhi
Gary	Kirsten	Jacques	Kallis	Cape Town
Gary	Kirsten	Herschelle	Gibbs	Cape Town
Jacques	Kallis	Herschelle	Gibbs	Cape Town

9. [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

Query:

```
select odi1.MatchID as match1,odi2.MatchID as match2,odi1.ground as Ground,odi1.date as Date1,odi2.Date as Date2
from odimatch odi1
inner join odimatch odi2 on odi2.ground = odi1.ground
where odi1.Date <> odi2.Date
and odi1.date < odi2.Date;
```

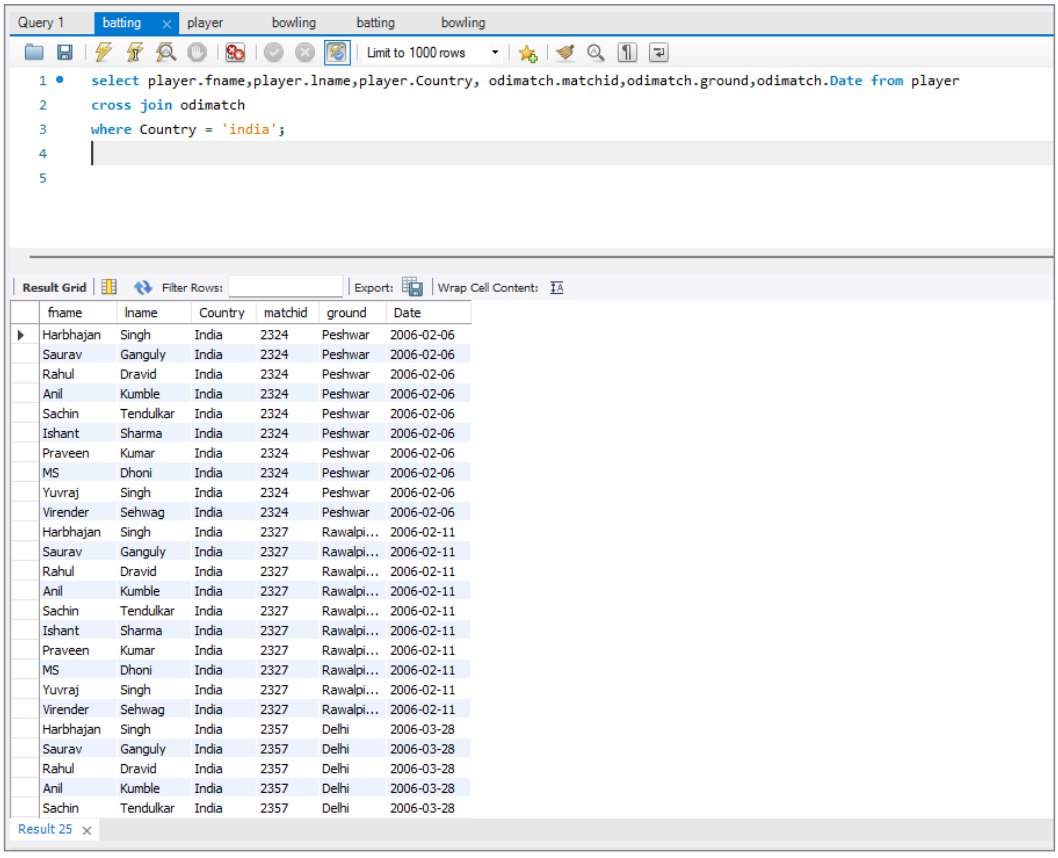


match1	match2	Ground	Date1	Date2
2404	2750	Colombo	2006-08-16	2008-08-24
2404	2755	Colombo	2006-08-16	2008-08-24

10. [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

Query:

```
select player.fname,player.lname,player.Country, odimatch.matchid,odimatch.ground,odimatch.Date from player
cross join odimatch
where Country = 'india';
```

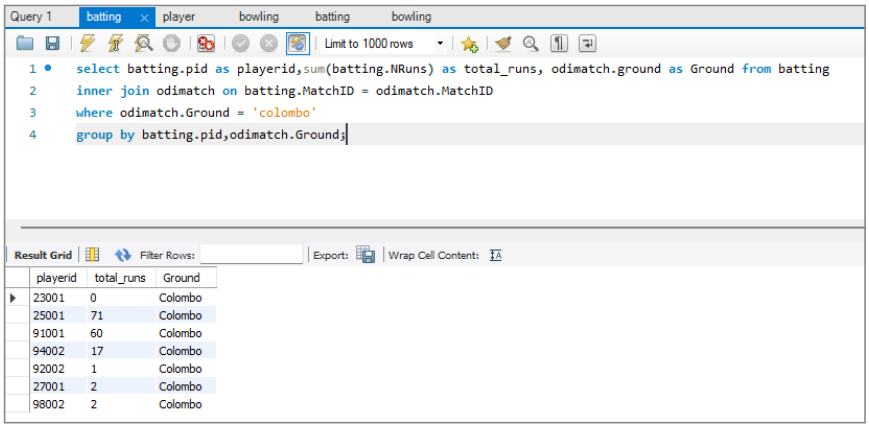


fname	lname	Country	matchid	ground	Date
Harbhajan	Singh	India	2324	Peshwar	2006-02-06
Saurav	Ganguly	India	2324	Peshwar	2006-02-06
Rahul	Dravid	India	2324	Peshwar	2006-02-06
Anil	Kumble	India	2324	Peshwar	2006-02-06
Sachin	Tendulkar	India	2324	Peshwar	2006-02-06
Ishant	Sharma	India	2324	Peshwar	2006-02-06
Praveen	Kumar	India	2324	Peshwar	2006-02-06
MS	Dhoni	India	2324	Peshwar	2006-02-06
Yuvraj	Singh	India	2324	Peshwar	2006-02-06
Virender	Sehwag	India	2324	Peshwar	2006-02-06
Harbhajan	Singh	India	2327	Rawalpi...	2006-02-11
Saurav	Ganguly	India	2327	Rawalpi...	2006-02-11
Rahul	Dravid	India	2327	Rawalpi...	2006-02-11
Anil	Kumble	India	2327	Rawalpi...	2006-02-11
Sachin	Tendulkar	India	2327	Rawalpi...	2006-02-11
Ishant	Sharma	India	2327	Rawalpi...	2006-02-11
Praveen	Kumar	India	2327	Rawalpi...	2006-02-11
MS	Dhoni	India	2327	Rawalpi...	2006-02-11
Yuvraj	Singh	India	2327	Rawalpi...	2006-02-11
Virender	Sehwag	India	2327	Rawalpi...	2006-02-11
Harbhajan	Singh	India	2357	Delhi	2006-03-28
Saurav	Ganguly	India	2357	Delhi	2006-03-28
Rahul	Dravid	India	2357	Delhi	2006-03-28
Anil	Kumble	India	2357	Delhi	2006-03-28
Sachin	Tendulkar	India	2357	Delhi	2006-03-28

11. [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

Query:

```
select batting.pid as playerid,sum(batting.NRuns) as total_runs, odimatch.ground as Ground from batting
inner join odimatch on batting.MatchID = odimatch.MatchID
where odimatch.Ground = 'colombo'
group by batting.pid,odimatch.Ground;
```

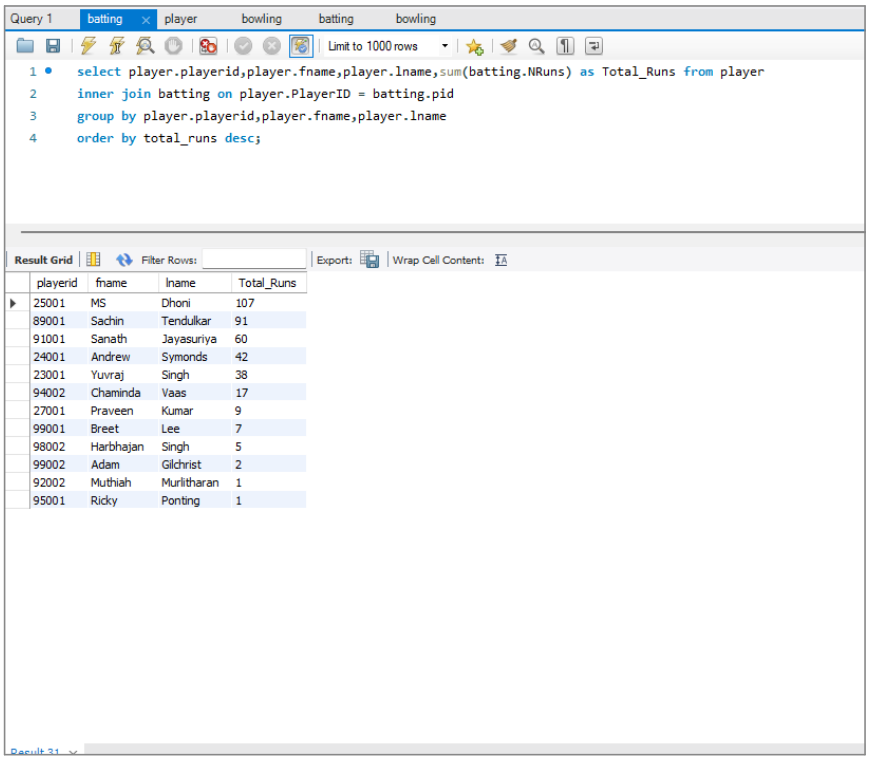


playerid	total_runs	Ground
23001	0	Colombo
25001	71	Colombo
91001	60	Colombo
94002	17	Colombo
92002	1	Colombo
27001	2	Colombo
98002	2	Colombo

12. [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

Query:

```
select player.playerid,player.fname,player.lname,sum(batting.NRuns) as Total_Runs from player
inner join batting on player.PlayerID = batting.pid
group by player.playerid,player.fname,player.lname
order by total_runs desc;
```

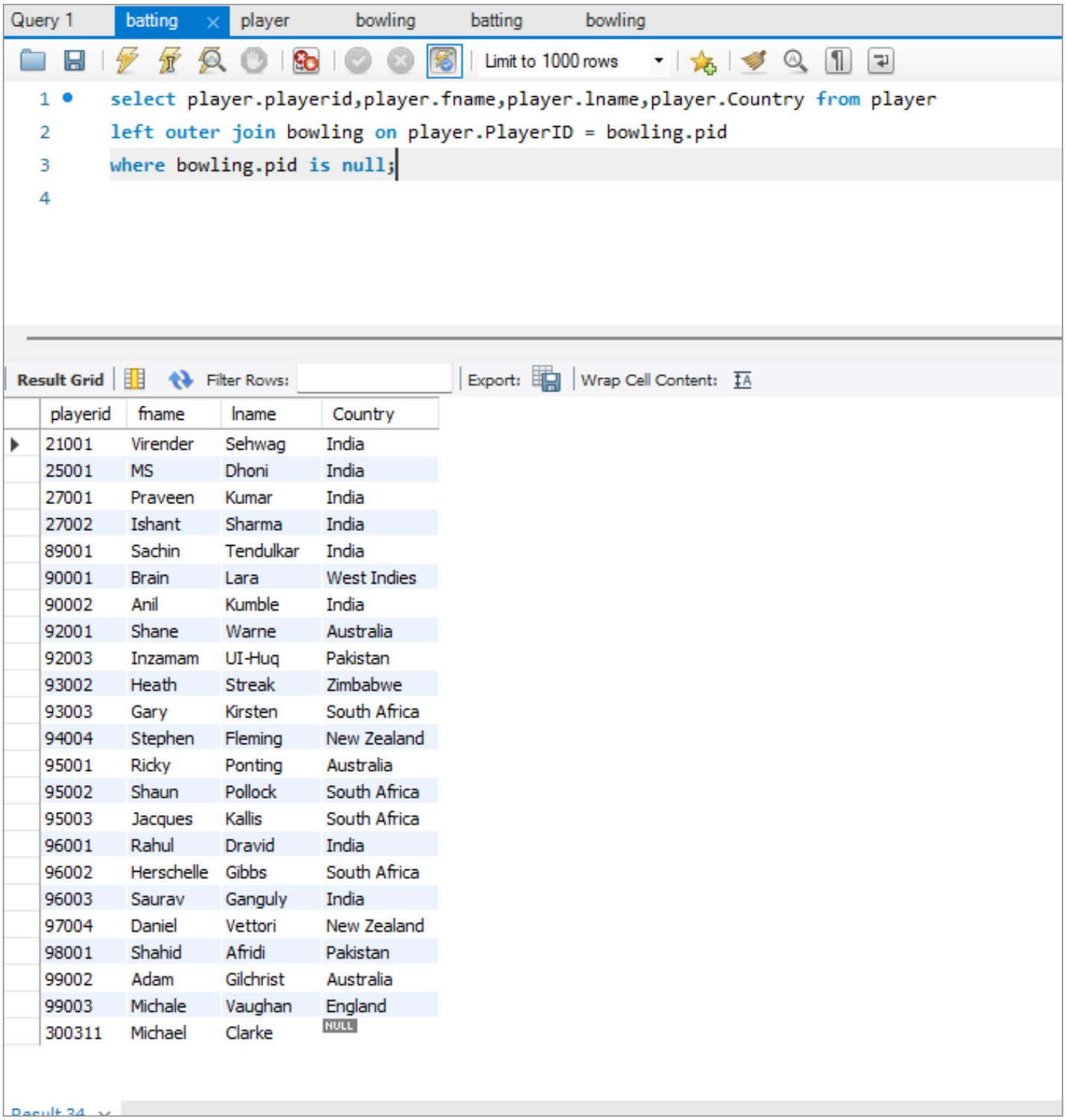


playerid	fname	lname	Total_Runs
25001	MS	Dhoni	107
89001	Sachin	Tendulkar	91
91001	Sanath	Jayasuriya	60
24001	Andrew	Symonds	42
23001	Yuvraj	Singh	38
94002	Chaminda	Vaas	17
27001	Praveen	Kumar	9
99001	Breet	Lee	7
98002	Harbhajan	Singh	5
99002	Adam	Gilchrist	2
92002	Muthiah	Murlitharan	1
95001	Ricky	Ponting	1

13. [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

Query:

```
select player.playerid,player.fname,player.lname,player.Country from player
left outer join bowling on player.PlayerID = bowling.pid
where bowling.pid is null;
```



playerid	fname	lname	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
300311	Michael	Clarke	NULL